

the HF index and actually does better. Over their sample period of 1996 to 2010, HFs returned 6.3% in excess of T-bills while the derivative strategy had excess returns of 10.2%.

Since HFs are short volatility, it is ironic that many asset owners, especially pension funds, purchase volatility protection while they own HFs that have the exact opposite exposure. They pay twice (handsomely, usually) to get back to the market portfolio.

3.6. LEVERAGE

One of the defining characteristics of HFs is that they use leverage, and, within the asset management industry, HFs make the most use of leverage. HFs use leverage to target a level of return volatility desired by investors, particularly for fixed income HFs where the underlying securities are (usually) not very volatile. HFs adjust their leverage in respond to time-varying investment opportunities. And finally, HFs lever to enhance returns (or, unfortunately, losses) on strategies that aren't sufficiently profitable on an unlevered basis.

Ang, Gorovyy, and van Inwegen (2011) examine HF leverage. When we wrote that paper, HFs did not publicly report their leverage, and ours was the first study to analyze HF leverage using actual leverage ratios. Now that the SEC gets reports on HF leverage, regulators will have a better picture of HF borrowing (even if investors might not).

We found that HF leverage is not that high—on average. There are some HFs (in fixed income) with leverage ratios above 30, but the overall HF industry has a leverage ratio around 2 or 3, as shown in Figure 17.8 (the solid line with the right-hand axis). This is because half the HF industry concentrates on equities, and leverage of equity HFs is, on average, modest. A more amazing fact is that HF leverage is counter-cyclical to the leverage of investment banks. I found this surprising because, prior to writing the paper, I thought HFs would be mini versions of investment banks in terms of leverage policies.

Figure 17.8 overlays the leverage of commercial banks, investment banks (the largest of which turned into commercial banks during the financial crisis), and the finance sector (plotted on the left-hand axis). Prior to 2007, HF leverage was around 2.3. HF leverage started to come down in 2007, a year before the financial crisis, and bottomed out below 1.5 in the last quarter of 2008 and the first quarter of 2009—the worst months of the crisis. Investment bank leverage has exactly the opposite pattern: as equity prices start to fall in 2008, leverage increases and peaks at over 40 right when HF leverage was lowest.

3.7. SUMMARY

HFs are a hodgepodge of factor risks—especially equity and volatility risk. Most HFs are short volatility and generate returns that are dependable most of the time